

When Does Aggregating Explanations Work?

TissueMNIST Results

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This report is generated from the structured result files listed in `RESULT_MANIFEST.json`. It is organised by model and experiment and retains incomplete snapshots together with additional result tables.

Additional result tables

A DenseNet-121

Table 1: TissueMNIST–DenseNet-121 results.

dataset	model	method	result status	gate status	passed	deterministic repeat	finite fraction	target sensitive	selected batch size	attack batch size	attack seconds per sample	attack peak cuda bytes	error
TissueMNIST	DenseNet-121	Saliency	passed	blocked	false	false	1		1				RuntimeError: Deterministic repeat check failed
TissueMNIST	DenseNet-121	InputXGradient	passed	blocked	false	false	1		1				RuntimeError: Deterministic repeat check failed
TissueMNIST	DenseNet-121	Integrated Gradients	passed	blocked	true	true	1	true	1	1	0.091962	6.6774e+09	
TissueMNIST	DenseNet-121	GuidedBackprop	passed	blocked	true	true	1	true	1				
TissueMNIST	DenseNet-121	Deconvolution	passed	blocked	false	false	1		1				RuntimeError: Deterministic repeat check failed
TissueMNIST	DenseNet-121	FeatureAbundance	passed	blocked	true	true	1	true	1				
TissueMNIST	DenseNet-121	Occlusion	passed	blocked	true	true	1	true	1				
TissueMNIST	DenseNet-121	DeepLift	passed	blocked	true	true	1	true	1	1	0.058259	6.5947e+08	
TissueMNIST	DenseNet-121	GradientShap	passed	blocked	true	true	1	true	1				
TissueMNIST	DenseNet-121	DeepLiftShap	passed	blocked	true	true	1	true	1	1	0.052809	1.7365e+09	
TissueMNIST	DenseNet-121	LRP	passed	blocked	true	true	1	true	1				

B EfficientNet-B0

Table 2: TissueMNIST-EfficientNet-B0 results.

dataset	model	method	result status	gate status	passed	deterministic repeat	finite fraction	target sensitive	selected batch size	attack batch size	attack seconds per sample	attack peak cuda bytes	error
TissueMNIST-B0	EfficientNet-B0	Saliency	failed	blocked	true	true	1	true	1	1	0.252425	3.8024e+08	RuntimeError: No explanation batch candidate passed
TissueMNIST-B0	EfficientNet-B0	InputXGradient	failed	blocked	true	true	1	true	1	1	0.181233	3.8265e+08	
TissueMNIST-B0	EfficientNet-B0	Integrated Gradients	failed	blocked	true	true	1	true	1	1	0.037191	4.5561e+09	
TissueMNIST-B0	EfficientNet-B0	GuidedBackprop	failed	blocked	true	true	1	true	1				
TissueMNIST-B0	EfficientNet-B0	Deconvolution	failed	blocked	true	true	1	true	1				
TissueMNIST-B0	EfficientNet-B0	FeatureAbundance	failed	blocked	true	true	1	true	1				
TissueMNIST-B0	EfficientNet-B0	Occlusion	failed	blocked	true	true	1	true	1				
TissueMNIST-B0	EfficientNet-B0	DeepLift	failed	blocked	true	true	1	true	1	1	0.023244	2.7867e+08	
TissueMNIST-B0	EfficientNet-B0	GradientShapley	failed	blocked	true	true	1	true	1				
TissueMNIST-B0	EfficientNet-B0	DeepLiftShapley	failed	blocked	true	true	1	true	1	1	0.022673	6.5413e+08	
TissueMNIST-B0	EfficientNet-B0	LRP	failed	blocked	false								

C MobileNetV3-Large

Table 3: TissueMNIST–MobileNetV3-Large results.

dataset	model	method	result status	gate status	passed	deterministic repeat	finite fraction	target sensitive	selected batch size	attack batch size	attack seconds per sample	attack peak cuda bytes	error
TissueMNIST-Large	MobileNetV3-Large	Sparsity	failed	blocked	true	true	1	true	1				RuntimeError: No explanation batch candidate passed
TissueMNIST-Large	MobileNetV3-Large	InputXGradient	failed	blocked	true	true	1	true	1				
TissueMNIST-Large	MobileNetV3-Large	Integrated Gradients	failed	blocked	true	true	1	true	1	1	0.024564	2.4103e+09	
TissueMNIST-Large	MobileNetV3-Large	GuidedBackprop	failed	blocked	true	true	1	true	1				
TissueMNIST-Large	MobileNetV3-Large	Deconvolution	failed	blocked	true	true	1	true	1				
TissueMNIST-Large	MobileNetV3-Large	Feature Ablation	failed	blocked	true	true	1	true	1				
TissueMNIST-Large	MobileNetV3-Large	Oclusion	failed	blocked	true	true	1	true	1				
TissueMNIST-Large	MobileNetV3-Large	DeepLift	failed	blocked	true	true	1	true	1	1	0.020956	2.2835e+08	
TissueMNIST-Large	MobileNetV3-Large	Gradient Shapley	failed	blocked	true	true	1	true	1				
TissueMNIST-Large	MobileNetV3-Large	DeepLift Shapley	failed	blocked	true	true	1	true	1	1	0.021406	5.0209e+08	
TissueMNIST-Large	MobileNetV3-Large	LRP	failed	blocked	false								

D ResNet-18

Table 4: TissueMNIST–ResNet-18 results.

dataset	model	method	result status	gate status	passed	deterministic repeat	finite fraction	target sensitive	selected batch size	attack batch size	attack seconds per sample	attack peak cuda bytes	error
TissueMNIST	ResNet-18	Saliency	passed	blocked	false	false	1		1				RuntimeError: Deterministic repeat check failed
TissueMNIST	ResNet-18	InputXGradient	passed	blocked	false	false	1		1				RuntimeError: Deterministic repeat check failed
TissueMNIST	ResNet-18	IntegratedGradients	passed	blocked	true	true	1	true	1	1	0.035272	1.2722e+09	RuntimeError: Deterministic repeat check failed
TissueMNIST	ResNet-18	GuidedBackprop	passed	blocked	false	false	1		1				RuntimeError: Deterministic repeat check failed
TissueMNIST	ResNet-18	Deconvolution	passed	blocked	false	false	1		1				RuntimeError: Deterministic repeat check failed
TissueMNIST	ResNet-18	FeatureAbandonment	passed	blocked	true	true	1	true	1				RuntimeError: Deterministic repeat check failed
TissueMNIST	ResNet-18	Occlusion	passed	blocked	true	true	1	true	1				RuntimeError: Deterministic repeat check failed
TissueMNIST	ResNet-18	DeepLift	passed	blocked	false	false	1		1				RuntimeError: Deterministic repeat check failed
TissueMNIST	ResNet-18	GradientShap	passed	blocked	true	true	1	true	1				RuntimeError: Deterministic repeat check failed
TissueMNIST	ResNet-18	DeepLiftShap	passed	blocked	true	true	1	true	1	1	0.016902	4.4585e+08	RuntimeError: Deterministic repeat check failed
TissueMNIST	ResNet-18	LRP	passed	blocked	true	true	1	true	1				RuntimeError: Deterministic repeat check failed

E ResNet-50

Table 5: TissueMNIST-ResNet-50 results.

dataset	model	method	result status	gate status	passed	deterministic repeat	finite fraction	target sensitive	selected batch size	attack batch size	attack seconds per sample	attack peak cuda bytes	error
TissueMNIST	ResNet-50	Saliency	passed	blocked	false	false	1		1				RuntimeError: Deterministic repeat check failed
TissueMNIST	ResNet-50	InputXGradient	passed	blocked	false	false	1		1				RuntimeError: Deterministic repeat check failed
TissueMNIST	ResNet-50	Integrated Gradients	passed	blocked	true	true	1	true	1	1	0.073951	4.5233e+09	RuntimeError: Deterministic repeat check failed
TissueMNIST	ResNet-50	GuidedBackprop	passed	blocked	false	false	1		1				RuntimeError: Deterministic repeat check failed
TissueMNIST	ResNet-50	Deconvolution	passed	blocked	false	false	1		1				RuntimeError: Deterministic repeat check failed
TissueMNIST	ResNet-50	FeatureAbandonment	passed	blocked	true	true	1	true	1				RuntimeError: Deterministic repeat check failed
TissueMNIST	ResNet-50	Occlusion	passed	blocked	true	true	1	true	1				RuntimeError: Deterministic repeat check failed
TissueMNIST	ResNet-50	DeepLift	passed	blocked	false	false	1		1				RuntimeError: Deterministic repeat check failed
TissueMNIST	ResNet-50	GradientShap	passed	blocked	true	true	1	true	1				RuntimeError: Deterministic repeat check failed
TissueMNIST	ResNet-50	DeepLiftShap	passed	blocked	true	true	1	true	1	1	0.026137	1.2474e+09	RuntimeError: Deterministic repeat check failed
TissueMNIST	ResNet-50	LRP	passed	blocked	true	true	1	true	1				RuntimeError: Deterministic repeat check failed

F ViT-B/16

Table 6: TissueMNIST–ViT-B/16 results.

dataset	model	method	result status	gate status	passed	deterministic repeat	finite fraction	target sensitive	selected batch size	attack batch size	attack seconds per sample	attack peak cuda bytes	error
TissueMNIST	ViT-B/16	Saliency	passed	passed	true	true	1	true	1				
TissueMNIST	ViT-B/16	InputXGradient	passed	passed	true	true	1	true	1				
TissueMNIST	ViT-B/16	IntegratedGradients	passed	passed	true	true	1	true	1	1	0.140803	6.6673e+09	
TissueMNIST	ViT-B/16	FeatureAbstraction	passed	passed	true	true	1	true	1				
TissueMNIST	ViT-B/16	Occlusion	passed	passed	true	true	1	true	1				
TissueMNIST	ViT-B/16	GradientShap	passed	passed	true	true	1	true	1				
TissueMNIST	ViT-B/16	CheferTransformerAttention	passed	passed	true	true	1	true	1				
TissueMNIST	ViT-B/16	PartialLRP	passed	passed	true	true	1	true	1				
TissueMNIST	ViT-B/16	FullLRP	passed	passed	true	true	1	true	1				
TissueMNIST	ViT-B/16	GradientAttentionRollout	passed	passed	true	true	1	true	1	1	0.062294	3.3842e+09	
TissueMNIST	ViT-B/16	AttentionGradCAM	passed	passed	true	true	1	true	1	1	0.051687	3.3105e+09	